Statistical Foundations of Machine Learning

Fast-Paced Tutorial

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26 September 2026

Q1. Topic: Inclusion–Exclusion and Conditional Probability

Let A, B, and C be events such that

$$P(A) = 0.20, \quad P(B) = 0.15, \quad P(C) = 0.10,$$
$$P(A \cap B) = 0.05, \quad P(A \cap C) = 0.04, \quad P(B \cap C) = 0.02,$$

and $P(A \cap B \cap C) = 0.01$.

(a) Find $P(A \cup B \cup C)$.

(b) Find the union-bound upper bound for $P(A \cup B \cup C)$.

(c) Find the second-order Bonferroni lower bound.

(d) Find $P(A \mid A \cup B \cup C)$.

Q2. Topic: Counting, PMF, CDF, Expectation and Variance

A box contains 5 nondefective items and 3 defective items. Three items are selected uniformly at random without replacement. Let X be the number of defective items selected.

(a) Find the PMF $p_X(k)$.

(b) Find the CDF $F_X(x)$.

(c) Calculate $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

Q3. Topic: Joint and Conditional PMFs

An urn contains 3 red, 4 white, and 5 blue balls. Three balls are selected uniformly at random without replacement. Let

$$X = \text{number of red balls selected}, \qquad Y = \text{number of white balls selected}.$$

(a) Find the joint PMF $p_{XY}(x, y)$.

(b) Find the marginal PMFs $p_X(x)$ and $p_Y(y)$.

(c) Find $P(X = 1 \mid Y = 1)$.

(d) Determine whether X and Y are independent.

Q4. Topic: Joint and Marginal PDFs

The joint PDF of X and Y is

$$f_{XY}(x, y) = \begin{cases} 2, & 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Verify that f_{XY} is a valid joint PDF.

(b) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.

(c) Find $f_{Y|X}(y \mid x)$.

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(d) Find

$$P\left(Y < \frac{X}{2}\right).$$

(e) Determine whether X and Y are independent.

Q5. Topic: Joint, Marginal and Conditional PDFs

Suppose

$$f_{XY}(x,y) = \begin{cases} c(x+y), & 0 < x < 1,\ 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the constant c.

(b) Find $f_X(x)$ and $f_Y(y)$.

(c) Find $f_{X|Y}(x \mid y)$.

(d) Find $P(X + Y \leq 1)$.

(e) Determine whether X and Y are independent.

Q6. Topic: Joint CDF and Independence

The joint CDF of X and Y is

$$F_{XY}(x,y) = \begin{cases} (1 - e^{-x})(1 - e^{-2y}), & x \geq 0,\ y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the joint PDF $f_{XY}(x,y)$.

(b) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.

(c) Determine whether X and Y are independent.

(d) Find $f_{Y|X}(y \mid x)$.

(e) Find $P(X < Y)$.

Q7. Topic: Covariance and Correlation

Suppose

$$\mathbb{E}[X] = 1, \qquad \mathbb{E}[Y] = 2, \qquad \text{Var}(X) = 4, \qquad \text{Var}(Y) = 9,$$

and

$$\text{Cov}(X,Y) = 3.$$

(a) Find $\text{Corr}(X,Y)$.

(b) Find $\mathbb{E}[2X - Y]$.

(c) Find $\text{Var}(2X - Y)$.

(d) Find $\text{Cov}(X + Y, X - Y)$.

(e) Are $X + Y$ and $X - Y$ uncorrelated?

Q8. Topic: Zero Correlation and Independence

Let

$$X \sim \text{Unif}(-1,1), \qquad Y = X^2.$$

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(a) Find $\mathbb{E}[X]$, $\mathbb{E}[Y]$, and $\mathbb{E}[XY]$.

(b) Find $\mathrm{Cov}(X, Y)$ and $\mathrm{Corr}(X, Y)$.

(c) Determine whether X and Y are independent.

(d) What does this example show about zero correlation and independence?

Q9. Topic: Covariance and Correlation

A researcher records the weekly study time X, measured in hours, and the examination score Y, measured out of 100, for a group of students. The following values are obtained:

$$\mathbb{E}[X] = 5, \qquad \mathbb{E}[Y] = 70, \qquad \mathbb{E}[XY] = 362,$$

and

$$\mathrm{Var}(X) = 4, \qquad \mathrm{Var}(Y) = 100.$$

(a) Find $\mathrm{Cov}(X, Y)$.

(b) Find $\mathrm{Corr}(X, Y)$.

(c) Interpret the sign and value of the correlation.

(d) Let $U = 60X$ be the study time measured in minutes. Find $\mathrm{Cov}(U, Y)$ and $\mathrm{Corr}(U, Y)$.

(e) Does changing the unit of study time from hours to minutes change the correlation?

Q10. Topic: Jointly Normal Variables

Suppose

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix} \right).$$

Define

$$U = X + Y, \qquad V = X - Y.$$

(a) Find the marginal distributions of X and Y.

(b) Find the distribution of $U = X + Y$.

(c) Find the distribution of $V = X - Y$.

(d) Determine whether U and V are independent. Justify your answer using the jointly normal property.

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Q1. Topic: Inclusion–Exclusion and Conditional Probability

Let A, B, and C be events such that

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$$P(A \cap B) = 0.05, \quad P(A \cap C) = 0.04, \quad P(B \cap C) = 0.02,$$

and $P(A \cap B \cap C) = 0.01$.

(a) Find $P(A \cup B \cup C)$.

(b) Find the union-bound upper bound for $P(A \cup B \cup C)$.

(c) Find the second-order Bonferroni lower bound.

(d) Find $P(A \mid A \cup B \cup C)$.

Solution

(a) $P(A \cup B \cup C)$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

$$= 0.20 + 0.15 + 0.10 - 0.05 - 0.04 - 0.02 + 0.01$$

$$= 0.35$$

(b) Find the union-bound upper bound for $P(A \cup B \cup C)$.

$$P(A \cup B \cup C) \leq P(A) + P(B) + P(C)$$

$$= 0.20 + 0.15 + 0.10$$

$$= 0.45$$

$$P(A \cup B \cup C) \leq 0.45$$

(c) Find the second-order Bonferroni lower bound.

$$P(A \cup B \cup C) \geq P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C)$$

Second order Bonferroni inequality

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Hence

$$P(A \cup B \cup C) \geq 0.45 - 0.11 = 0.34$$

$$\boxed{P(A \cup B \cup C) \geq 0.34}$$

(d) Find $P(A \mid A \cup B \cup C)$.

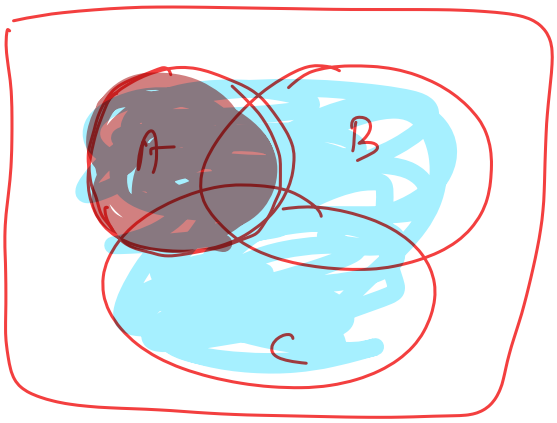
$$P(A \mid A \cup B \cup C) = \frac{P(A \cap (A \cup B \cup C))}{P(A \cup B \cup C)}$$

$$= \frac{P(A)}{P(A \cup B \cup C)}$$

$$= \frac{0.20}{0.35} = \frac{4}{7}$$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

provided $P(B) > 0$



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Q2. Topic: Counting, PMF, CDF, Expectation and Variance Hypergeometric distribution

A box contains 5 nondefective items and 3 defective items. Three items are selected uniformly at random without replacement. Let X be the number of defective items selected.

(a) Find the PMF $p_X(k)$.
(b) Find the CDF $F_X(x)$.
(c) Calculate $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

Total items = 5 + 3 = 8 (5: nondefective, 3: defective)

If $X = k$ Then we must k defective items and 3-k non-defective items

$k \in \{0, 1, 2, 3\}$

Probability of choosing k defective items

$$= \frac{\text{number of selection with exactly k defective items}}{\text{total number of selections}}$$

$$= \frac{\binom{3}{k}\binom{5}{3-k}}{\binom{8}{3}}$$

(a) pmf of $p_X(k)$

$$p_X(k) = P(X=k) = \frac{\binom{3}{k}\binom{5}{3-k}}{\binom{8}{3}} \qquad k \in \{0, 1, 2, 3\}$$

$$= \begin{cases} \frac{5}{28} & ; k=0 \\ \frac{15}{28} & ; k=1 \\ \frac{15}{56} & ; k=2 \\ \frac{1}{56} & ; k=3 \\ 0 & ; \text{otherwise} \end{cases}$$

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(b) CDF $F_X(x)$

$$F_X(x) = P(X \le x)$$

$$= \begin{cases} 0 & ; x < 0 \\ \frac{5}{28} & 0 \le x < 1 \\ \frac{5}{28} + \frac{15}{28} = \frac{5}{7} & 1 \le x < 2 \\ \frac{5}{7} + \frac{15}{56} = \frac{55}{56} & 2 \le x < 3 \\ 1 & x \ge 3 \end{cases}$$

(c) $E[X]$ and $Var(X)$

$$E[X] = \sum_{k \in S_X} k\, p_X(k)$$

$$Var(X) = E[X^2] - (E[X])^2$$

$$E[X] = \sum_{k=0}^{3} k\, p_X(k)$$

$$= 0\, p_X(0) + 1.\, p_X(1) + 2\, p_X(2) + 3\, p_X(3)$$

$$= 0 + \frac{30}{56} + 2.\frac{15}{56} + 3.\frac{1}{56} = \boxed{\frac{9}{8}}$$

$$E[X^2] = \sum_{k=0}^{3} k^2 p_X(k) = 0^2 p_X(0) + 1^2 p_X(1) + 2^2 p_X(2) + 3^2 p_X(3)$$

$$= 0 + \frac{30}{56} + 4.\frac{15}{56} + 9.\frac{1}{56} = \frac{99}{56}$$

$$Var(X) = \frac{99}{56} - \left(\frac{9}{8}\right)^2 = \boxed{\frac{225}{448}}$$

: Think of Hypergeometric distributions

$X \sim \text{Hypergeometric}(N=8, K=3, n=3)$

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Q3. Topic: Joint and Conditional PMFs

An urn contains 3 red, 4 white, and 5 blue balls. Three balls are selected uniformly at random without replacement. Let

$$X = \text{number of red balls selected}, \qquad Y = \text{number of white balls selected}.$$

(a) Find the joint PMF $p_{XY}(x, y)$.

(b) Find the marginal PMFs $p_X(x)$ and $p_Y(y)$.

(c) Find $P(X = 1 \mid Y = 1)$.

(d) Determine whether X and Y are independent.

Total balls = 3R + 4W + 5B = 12 balls

$X = x$ and $Y = y$ means we must select x red balls, y white balls and $3-x-y$ blue balls.

(a) Joint PMF $p_{XY}(x,y)$

$$p_{XY}(x,y) = P(X=x, Y=y) = \begin{cases} \dfrac{\binom{3}{x}\binom{4}{y}\binom{5}{3-x-y}}{\binom{12}{3}} & x, y \geq 0,\ x+y \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

$p_{X,Y}(x,y)$	$y=0$	$y=1$	$y=2$	$y=3$	
$x=0$	$\frac{10}{220}$ $= p_{XY}(0,0)$	$\frac{40}{220}$	$\frac{30}{220}$	$\frac{4}{220}$	$p_X(0)$
$x=1$	$\frac{30}{220}$	$\frac{60}{220}$	$\frac{18}{220}$	0	$p_X(1)$
$x=2$	$\frac{15}{220}$	$\frac{12}{220}$	0	0	$p_X(2)$
$x=3$	$\frac{1}{220}$	0	0	0	$p_X(3)$

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(b) Marginal PMF $p_X(x)$ and $p_Y(y)$

Marginal of X → Sum over y

$$p_X(0) = \frac{84}{220} = \frac{21}{55}$$ ✓

$$p_X(1) = \frac{108}{220} = \frac{27}{55}$$ ✓

$$p_X(2) = \frac{27}{220}$$

$$p_X(3) = \frac{1}{220}$$ ✓

$$p_X(x) = \begin{cases} \frac{21}{55} & x = 0 \\ 27/55 & x = 1 \\ \frac{27}{220} & x = 2 \\ \frac{1}{220} & x = 3 \\ 0 & \text{otherwise} \end{cases}$$ ✓

Similarly:

$$p_Y(y) = \begin{cases} \frac{14}{55} & ; y = 0 \\ \frac{28}{55} & ; y = 1 \\ \frac{12}{55} & y = 2 \\ 1/55 & y = 3 \\ 0 & \text{else} \end{cases}$$

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(c) $P(X=1 \mid Y=1)$

$$= \frac{P(X=1, Y=1)}{P(Y=1)}$$

$$= \frac{P_{XY}(1,1)}{P_Y(1)} \quad \text{table}$$

$$= \frac{60}{220} \Big/ \frac{28 \times 4}{220} = \boxed{\frac{15}{28}}$$

(d) Indepence of X and Y.

Not independent

$P(X=3) > 0$

$P(Y=3) > 0$

$P(X=3, Y=3) = 0$

$X+Y \leq 3$

$3+3 = 6 \not\leq 3$

$\Rightarrow \boxed{P(X=3, Y=3) \neq P_X(3)\, P_Y(3)}$

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Q4. Topic: Joint and Marginal PDFs

The joint PDF of X and Y is

$$f_{XY}(x,y) = \begin{cases} 2, & 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Verify that f_{XY} is a valid joint PDF.

(b) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.

(c) Find $f_{Y|X}(y \mid x)$.

(d) Find

$$P\left(Y < \frac{X}{2}\right).$$

(e) Determine whether X and Y are independent.

Solutions

$f_{XY}(x,y) \geq 0$

$\iint f_{XY}(x,y)\,dx\,dy = 1$

(a) Positivity →

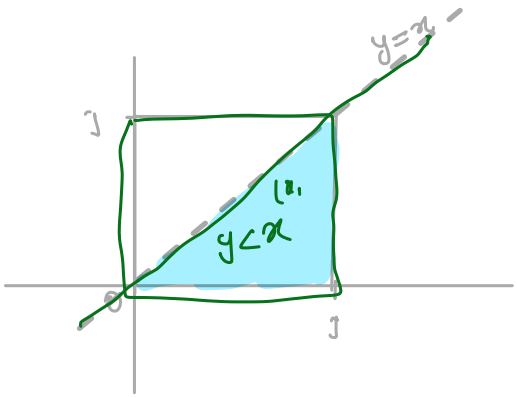
$$\int_0^1 \int_0^x 2\,dy\,dx$$

$$\int_0^x dy = [y]_0^x = x - 0$$

$$= \int_0^1 2x\,dx$$

$$= 2\left[\frac{x^2}{2}\right]_0^1 = 2\left(\frac{1}{2} - 0\right)$$

$$= 1 \quad \checkmark$$



(b) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,dy = \int_0^x 2\,dy = 2x \qquad 0<x<1$$

$$\therefore \quad f_X(x) = \begin{cases} 2x & 0<x<1 \\ 0 & \text{otherwise} \end{cases}$$

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$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,dx = \int_{x=y}^{1} 2\,dx = 2(1-y) \;;\; 0<y<1$$

$$f_Y(y) = \begin{cases} 2(1-y) & ; \; 0<y<1 \\ 0 & \text{otherwise} \end{cases}$$

(c) Find $f_{Y|X}(y \mid x)$.

$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)}$$

$$f_{XY}(x,y) = 2 \qquad 0<y<x<1$$

$$f_X(x) = 2x \qquad 0<x<1$$

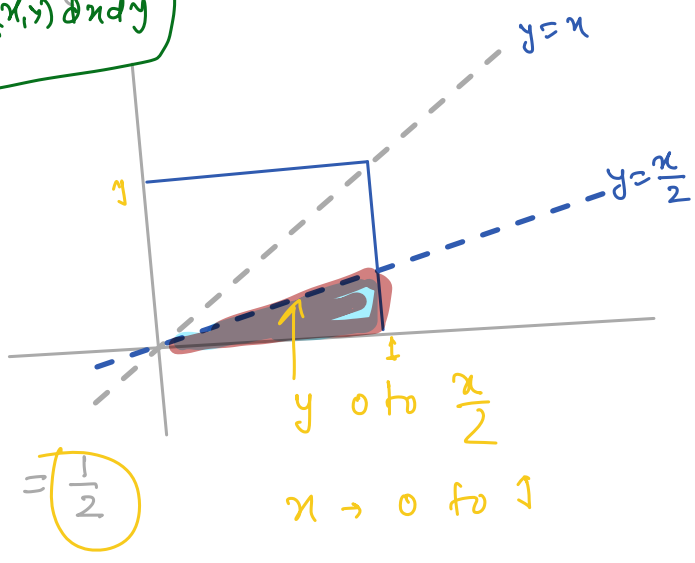
$$\therefore \quad f_{Y|X}(y|x) = \frac{2}{2x} = \begin{cases} \frac{1}{x} & ; \; 0<y<x \\ 0 & \text{otherwise} \end{cases}$$

$$P((x,y) \in B) = \iint_B f_{XY}(x,y)\,dx\,dy$$

(d) $P\left(Y < \frac{X}{2}\right)$.

$$= \int_0^1 \int_{y=0}^{x/2} 2\,dy\,dx$$

$$= \int_0^1 2 \cdot \frac{x}{2}\,dx = \left(\frac{x^2}{2}\right)_0^1 = \frac{1}{2}$$



(e) X and Y is not independent

→ Support is not disjoint

Also check $f_{XY}(0.2, 0.8) \neq f_X(0.2)\, f_X(0.8)$

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Q5. Topic: Joint, Marginal and Conditional PDFs

Suppose

$$f_{XY}(x,y) = \begin{cases} c(x+y), & 0 < x < 1,\ 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the constant c.

(b) Find $f_X(x)$ and $f_Y(y)$.

(c) Find $f_{X|Y}(x \mid y)$.

(d) Find $P(X+Y \le 1)$.

(e) Determine whether X and Y are independent.

$y < x$

↳ valid pdf $\iint f_{XY}\, dx\, dy = 1$

$f_{XY}(x,y) = f_X(x) f_Y(y)$

similarly → check with CDF

(a) Find the constant c.

$$\int_{y=0}^{1} \int_{x=0}^{1} c(x+y)\, dx\, dy = 1$$

$$c \int_{y=0}^{1} \left(\frac{x^2}{2} + yx\right)_0^1 dy = c \int_{y=0}^{1} \left(\frac{1}{2} + y\right) dy$$

$$= c \left(\frac{1}{2}y + \frac{y^2}{2}\right)_0^1$$

$$= c\left(\frac{1}{2} + \frac{1}{2} - 0\right) = c$$

$$\Rightarrow \boxed{c = 1}$$

(b) Find $f_X(x)$ and $f_Y(y)$.

$$f_X(x) = \int_{y=0}^{1} f_{XY}(x,y)\, dy$$

$$= \int_{y=0}^{1} (x+y)\, dy$$

$$= x + \frac{1}{2}$$

$$\Rightarrow f_X(x) = \begin{cases} x + \frac{1}{2} & ; 0 < x < 1 \\ 0 & \text{else} \end{cases}$$

Similarly $f_Y(y) = \begin{cases} y + \frac{1}{2} & ; 0 < y < 1 \\ 0 & \text{else} \end{cases}$

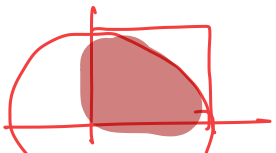
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(c) Find $f_{X|Y}(x \mid y)$.

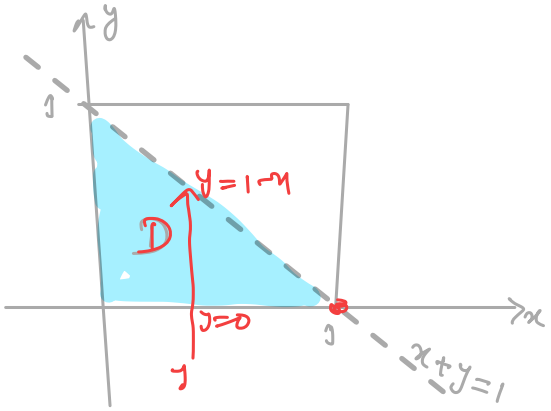
$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$= \begin{cases} \dfrac{x+y}{y+\frac{1}{2}} ; & 0<x<1,\ 0<y<1 \\ 0 & \text{else} \end{cases}$$

(d) Find $P(X+Y \le 1)$. $P(x^2+y^2<1)$



$$P(X+Y \le 1) = \iint_D f_{XY}(x,y)\,dx\,dy$$



$$= \int_{x=0}^{1} \int_{y=0}^{1-x} (x+y)\,dy\,dx$$

$$= \int_{x=0}^{1} \left[xy + \frac{y^2}{2} \right]_{y=0}^{1-x} dx = \int_{x=0}^{1} \left[x(1-x) + \frac{(1-x)^2}{2} \right] dx$$

$$= \int_{x=0}^{1} \left(\frac{1}{2} - \frac{x^2}{2} \right) dx$$

$$= \left(\frac{1}{2}x - \frac{x^3}{6} \right)_0^1 = \frac{1}{2} - \frac{1}{6} = \boxed{\frac{1}{3}}$$

(e) Determine whether X and Y are independent.

$$f_X(x)\, f_Y(y) = \left(x+\tfrac{1}{2}\right)\left(y+\tfrac{1}{2}\right) \neq (x+y)$$

$$\neq f_{XY}(x,y)$$

$\Rightarrow$ Not independent.

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Q6. Topic: Joint CDF and Independence

The joint CDF of X and Y is

$$F_{XY}(x,y) = \begin{cases} (1-e^{-x})(1-e^{-2y}), & x \geq 0,\ y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the joint PDF $f_{XY}(x,y)$.

(b) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.

(c) Determine whether X and Y are independent.

(d) Find $f_{Y|X}(y \mid x)$.

(e) Find $P(X < Y)$.

differentiate the CDF ↓ get pdf

(a) Find the joint PDF $f_{XY}(x,y)$.

$$f_{XY}(x,y) = \frac{\partial^2}{\partial x\, \partial y} F_{XY}(x,y)$$

→ Lecture 11

$$\frac{\partial}{\partial x} F_{XY}(x,y) = e^{-x}(1-e^{-2y})$$

$$\therefore f_{XY}(x,y) = \frac{\partial}{\partial y}\left(e^{-x}(1-e^{-2y})\right)$$

$$= 2e^{-x}e^{-2y}$$

$$= 2e^{-(x+2y)}$$

$$\therefore f_{XY}(x,y) = \begin{cases} 2e^{-(x+2y)}; & x>0,\ y>0 \\ 0 & \text{else} \end{cases}$$

(b) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.

$$F_X(x) = \lim_{y\to\infty} F_{XY}(x,y)$$

$$= \lim_{y\to\infty} (1-e^{-x})(1-e^{-2y})$$

$$= (1-e^{-x})$$

$$\Rightarrow f_X(x) = \frac{d}{dx}(1-e^{-x}) = e^{-x} \quad x>0$$

$$F_X(x) = \lim_{y\to\infty} F_{XY}(x,y)$$

$$F_Y(y) = \lim_{x\to\infty} F_{XY}(x,y)$$

$$f_X(x) = \frac{d}{dx} F_X(x)$$

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

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$$\Rightarrow \quad \boxed{f_X(x) = \begin{cases} e^{-x} & : x > 0 \\ 0 & \text{else} \end{cases}}$$

Similarly $F_Y(y) = \lim_{x\to\infty} F_{XY}(x,y) = \lim_{x\to\infty} (1-e^{-x})(1-e^{-2y})$

$$= 1 - e^{-2y}$$

$F_Y(y)$,,

$$f_Y(y) = \frac{d}{dy}(1-e^{-2y}) = 2e^{-2y}$$

$$\boxed{f_Y(y) = \begin{cases} 2e^{-2y} & ; y > 0 \\ 0 & \text{else} \end{cases}}$$

(c) Determine whether X and Y are independent.

$$f_X(x)\, f_Y(y) = e^{-x} 2e^{-2y}$$

$$= 2e^{-(x+2y)} = f_{XY}(x,y)$$

$\Rightarrow$ X and Y are independent.

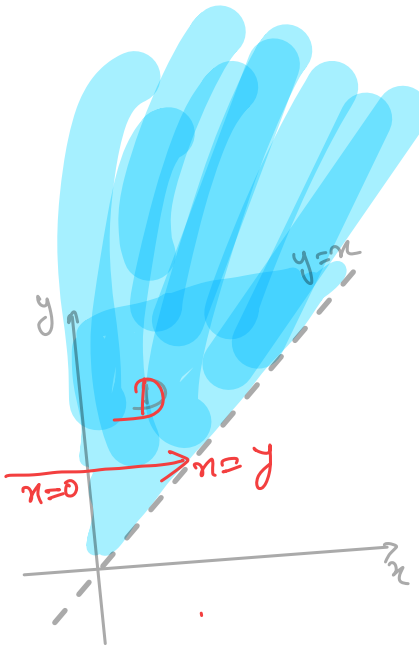
(d) Find $f_{Y|X}(y \mid x)$.

$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{2e^{-(x+2y)}}{e^{-x}} = 2e^{-2y} \quad : y > 0$$

$$\boxed{f_{Y|X}(y|x) = \begin{cases} 2e^{-2y} & : y > 0 \\ 0 & \text{else} \end{cases}}$$

(e) Find $P(X < Y)$.

$$= \iint_D f_{XY}(x,y)\, dx\, dy = \int_{y=0}^{\infty} \int_{x=0}^{y} 2e^{-(x+2y)}\, dx\, dy$$



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$$\int_{x=0}^{y} 2e^{-x-2y}\,dx = 2e^{-2y}\int_{x=0}^{y} e^{-x}\,dx$$

$$= 2e^{-2y}\left[-e^{-x}\right]_0^y$$

$$= 2e^{-2y}\left(-e^{-y}+1\right)$$

$$= 2e^{-2y}\left(1-e^{-y}\right)$$ ✓

$$\int_0^{\infty} 2e^{-2y}\left(1-e^{-y}\right)dy = 2\int_0^{\infty}\left(e^{-2y}-e^{-3y}\right)dy$$

$$= 2\left[\frac{e^{-2y}}{-2} - \frac{e^{-3y}}{-3}\right]_0^{\infty}$$

$$= 2\left[0 - \left(\frac{1}{-2}+\frac{1}{3}\right)\right]$$

$$= 2\left(\frac{1}{2}-\frac{1}{3}\right)$$

$$= 1-\frac{2}{3} = \frac{1}{3}$$ ✓

$\therefore$ $\boxed{P(X<Y) = \frac{1}{3}}$ ✓

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Q7. Topic: Covariance and Correlation

Suppose

$$\mathbb{E}[X] = 1, \qquad \mathbb{E}[Y] = 2, \qquad \mathrm{Var}(X) = 4, \qquad \mathrm{Var}(Y) = 9,$$

and

$$\mathrm{Cov}(X, Y) = 3.$$

(a) Find $\mathrm{Corr}(X, Y)$.
(b) Find $\mathbb{E}[2X - Y]$.
(c) Find $\mathrm{Var}(2X - Y)$.
(d) Find $\mathrm{Cov}(X + Y, X - Y)$.
(e) Are $X + Y$ and $X - Y$ uncorrelated?

$$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}$$

(a) $\mathrm{Corr}(X,Y)$

$$= \frac{3}{\sqrt{4 \times 9}}$$

$$= \frac{1}{2}$$

$$E[aX + bY] = a\,E[X] + b\,E[Y]$$

(b) $E[2X - Y]$

$$= 2\,E[X] - E[Y]$$

$$= 2 \cdot 1 - 2$$

$$= 0$$

(c) $\mathrm{Var}(2X - Y)$

$$\mathrm{Var}(aX + bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X,Y)$$

$$= 2^2\,\mathrm{Var}(X) + (-1)^2\,\mathrm{Var}(Y) + 2(2)(-1)\,\mathrm{Cov}(X,Y)$$

$$= 4(4) + 1(9) - 4(3)$$

$$= 13$$

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(d) $\text{Cov}(X+Y, X-Y)$

$$\text{Cov}(X+Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$$

$$\text{Cov}(aX, bY) = ab\,\text{Cov}(X, Y)$$

$$= \text{Cov}(X, X-Y) + \text{Cov}(Y, X-Y)$$

$$= \text{Cov}(X, X) - \text{Cov}(X, Y) + \text{Cov}(Y, X) - \text{Cov}(Y, Y)$$

$$= \text{Cov}(X, X) - \text{Cov}(Y, Y)$$

$$= \text{Var}(X) - \text{Var}(Y)$$

$$= 4 - 9$$

$$= -5$$

(e) Are $X + Y$ and $X - Y$ uncorrelated?

X, Y uncorrelated if $\text{Cov}(X, Y) = 0$

Since $\text{Cov}(X+Y, X-Y) = -5 \neq 0$

These are correlated.

Correlation $\neq 0 \Rightarrow$ C

If $\text{Corr}(X, Y) = 0$ then we say X and Y are uncorrelated

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Q8. Topic: Zero Correlation and Independence

Let

$$X \sim \text{Unif}(-1,1), \qquad Y = X^2.$$

$E\,g(X) = \int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx$, $y = g(x) = x^2$, $E(g(X))$

$a \; b$, $b - a = 1 - (-1) = 2$

(a) Find $\mathbb{E}[X]$, $\mathbb{E}[Y]$, and $\mathbb{E}[XY]$.

(b) Find $\text{Cov}(X,Y)$ and $\text{Corr}(X,Y)$.

(c) Determine whether X and Y are independent.

(d) What does this example show about zero correlation and independence?

$X \sim \text{Unif}(-1,1) \Rightarrow f_X(x) = \begin{cases} \frac{1}{2} & ; -1 < x < 1 \\ 0 & ; \text{else} \end{cases}$

a

$$E[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx = \frac{1}{2}\int_{-1}^{1} x\, dx = 0$$

$\left[\frac{x^2}{2}\right]_{-1}^{1} = \frac{1}{2} - \left(\frac{-1}{2}\right) = \frac{1}{2} - \frac{1}{2} = 0$

$$E[Y] = E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x)\, dx = \frac{1}{2}\int_{-1}^{1} x^2\, dx = \frac{1}{3}$$

$\left[\frac{x^3}{3}\right]_{-1}^{1} = \frac{1}{2} - \left(\frac{-1}{3}\right) = \frac{2}{3}$

$$E[XY] = E[X^3] = \int_{-\infty}^{\infty} x^3 f_X(x)\, dx = \frac{1}{2}\int_{-1}^{1} x^3\, dx = 0$$

$\left[\frac{x^4}{4}\right]_{-1}^{1}$

b

$$\text{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

$$\text{Cov}(X,Y) = E[XY] - E[X]\,E[Y]$$

$$= 0 - 0\left(\tfrac{1}{3}\right) = 0$$

$$\text{Corr}(X,Y) = \frac{\text{Cov}(X,Y)}{\sqrt{\text{var}(X)}\,\sqrt{\text{var}(Y)}} = 0$$

c X and Y are not independent — Why? $Y = X^2$

d Zero correlation does not imply independence.

$\text{Cov}(X,Y) = 0 = \text{Corr}(X,Y)$

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Q9. Topic: Covariance and Correlation

A researcher records the weekly study time X, measured in hours, and the examination score Y, measured out of 100, for a group of students. The following values are obtained:

$$\mathbb{E}[X] = 5, \qquad \mathbb{E}[Y] = 70, \qquad \mathbb{E}[XY] = 362,$$

and

$$\mathrm{Var}(X) = 4, \qquad \mathrm{Var}(Y) = 100.$$

(a) Find $\mathrm{Cov}(X, Y)$.

(b) Find $\mathrm{Corr}(X, Y)$.

(c) Interpret the sign and value of the correlation.

(d) Let $U = 60X$ be the study time measured in minutes. Find $\mathrm{Cov}(U, Y)$ and $\mathrm{Corr}(U, Y)$.

(e) Does changing the unit of study time from hours to minutes change the correlation?

$E(XY) - E(X)E(Y)$

$\mathrm{Corr}(X,Y) = \dfrac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}$

(a) $\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$

$= 362 - 70 \times 5$

$= 12$

(b) $\mathrm{Corr}(X,Y) = \dfrac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}}$

$-1 \le \rho \le 1$

$\rho = 1$

$= \dfrac{12}{2 \times 10} = \dfrac{3}{5} = +0.60$

(c) These are positively correlated

(d) $U = 60X$

$\mathrm{Cov}(aX, bY) = ab\,\mathrm{Cov}(X,Y)$

$\mathrm{Cov}(U,Y) = \mathrm{Cov}(60X, Y)$

$= 60\,\mathrm{Cov}(X,Y) = 60 \times 12 = 720$

$\mathrm{Corr}(U,Y) = \dfrac{\mathrm{Cov}(U,Y)}{\sqrt{\mathrm{Var}(U)\,\mathrm{Var}(Y)}}$

$\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$

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$$\text{Var}(U) = \text{Var}(60X) = (60)^2 \text{Var}(X)$$

$$= (60)^2 \cdot 4$$

$$\sqrt{\text{Var}(U)} = 60 \times 2 = 120$$

$$\therefore \text{Cov}(U, Y) = \frac{720}{120 \times 10} = \frac{60}{100} = 0.6$$

(e) Changing the units from hour to minutes does not change the correlation but changes the covariance.

Covariance : depends on unit of measurement
Correlation → Has no units

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Q10. **Topic: Jointly Normal Variables**

Suppose

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix} \right).$$

Define

$$U = X + Y, \qquad V = X - Y.$$

(a) Find the marginal distributions of X and Y.

(b) Find the distribution of $U = X + Y$.

(c) Find the distribution of $V = X - Y$.

(d) Determine whether U and V are independent. Justify your answer using the jointly normal property.

Remember this

- Joint distribution:

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}_2 \left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix} \right).$$

- Marginal distributions:

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \qquad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2).$$

- Covariance and correlation:

$$\mathrm{Cov}(X, Y) = \rho\sigma_X\sigma_Y, \qquad \mathrm{Corr}(X, Y) = \rho.$$

- Linear combination:

$$aX + bY \sim \mathcal{N}\left(a\mu_X + b\mu_Y,\ a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y\right)$$

$\mu_X = 0 \qquad \sigma_X^2 = 4 \qquad \mathrm{Cov}(X,Y) = 2$

$\mu_Y = 0 \qquad \sigma_Y^2 = 4 \qquad \mathrm{Corr}(X,Y) = \frac{2}{2 \cdot 2} = \frac{1}{2}$

(a) $X \sim N(0, 4)$

$Y \sim N(0, 4)$

(b) $U = X + Y$

$\mu_U = E[U] = E[X] + E[Y] = 0$

$\mathrm{Var}(U) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$

$= 4 + 4 + 2(2) = 12$

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$$U = X + Y \sim N(0, 12)$$

(c) $V = X - Y$

$$E[V] = E[X] - E[Y] = 0$$

$$\text{Var}(V) = \text{Var}(X) + (-1)^2 \text{Var}Y - 2.\text{Cov}(X, Y)$$

$$= 4 + 4 - 2.(2)$$

$$= 4$$

$$\therefore \quad V \sim N(0, 4)$$

(d) $\text{Cov}(U, V) = \text{Cov}(X + Y, X - Y)$

$$= \text{Cov}(X, X) + \text{Cov}(Y, X) - \text{Cov}(X, Y) - \text{Cov}(Y, Y)$$

$$= \text{Var}(X) - \text{Var}(Y)$$

$$= 4 - 4$$

$$= 0$$

Bernaudi

Normal

Covariance zero $\Rightarrow$ Independence.

$U \sim N(0, 12) \quad V \sim N(0, 4) \quad \text{Cov}(U, V) = 0$

$$\begin{pmatrix} U \\ V \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 12 & 0 \\ 0 & 4 \end{pmatrix} \right)$$

$\Rightarrow$ U and V are Independent

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